Lab 1 - Competition: Music taste prediction

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Data analysis

Dataset overview	Result
Original labeled songs	750
Additional exact duplicate records	14
Songs after deduplication	736
Development / holdout split	588 / 148
Input features	13: 10 numerical and 3 categorical
Missing values	0
Development-set labels	Like: 354 (60.20%); Dislike: 234 (39.80%)

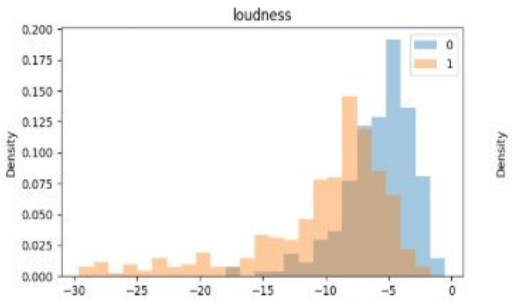
Records with identical features and labels were treated as duplicates, with one record retained from each group. This is an assumption made for this experiment.

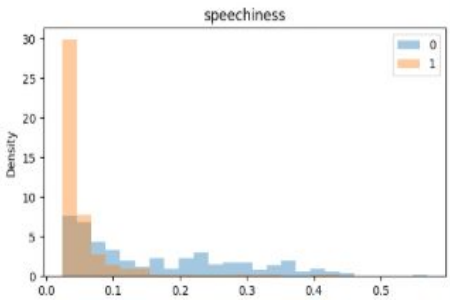
The development set was used for data exploration and model selection. The holdout set was reserved for evaluation after selection.

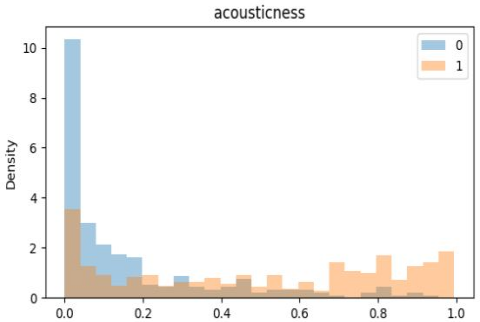
Data analysis

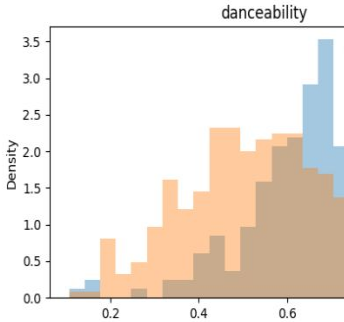
Numerical Feature Distributions

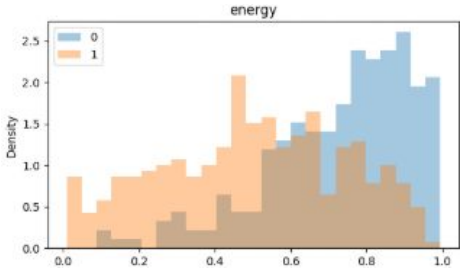
- In the development set, liked songs tend to have higher acousticness and lower energy, loudness (more negative values), and speechiness. danceability shows a smaller difference, with substantial overlap between the classes.









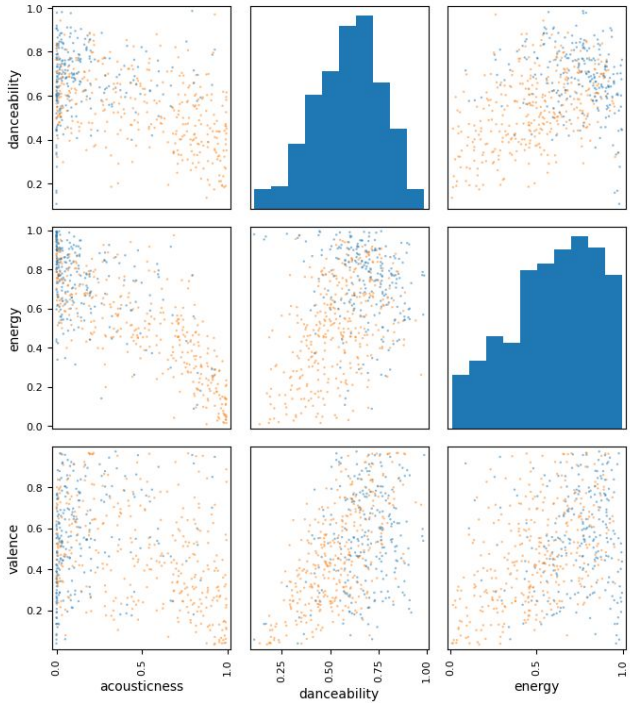


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Method Selection

Data observation

For example, both Like and Dislike songs occur at low acousticness values. The two classes also overlap in the danceability–energy scatter plot. A single feature does not clearly separate them, so I tested models that combine multiple features. The overlap is visible in the feature pairs shown here. These plots do not tell us whether the classes can be separated when other features are included.



Logistic Regression

Combines features to estimate the probability that a song is liked. I used it to tes
a relatively simple linear classifier and provide a comparison baseline.

Linear SVM

Finds a linear decision boundary with a large margin between classes. I used it t
compare a different way of learning a linear boundary with Logistic Regression.

Method Selection

Random Forest
Trains many decision trees using different samples and randomly selected features at each split, then combines their predictions. I used it to test whether flexible rules involving several features perform better than the linear models.

Gradient Boosting
Builds decision trees sequentially, with each new tree helping to reduce the errors of the current model. I used it to compare a different tree ensemble approach with Random Forest.

Feature Preprocessing

- **Numerical features:** duration is measured in milliseconds, while energy ranges from 0 to 1. I standardized numerical features for Logistic Regression and SVM so their different scales would not affect model training. Tree-based models used the original values because they split data using feature thresholds.
- **Categorical features:** key, mode, and time_signature are stored as numbers but represent categories. I used one-hot encoding so the models would not treat their numbers as a continuous scale.
- **Outlier check:** The 1.5 × IQR rule flagged 41 of the 588 development songs with unusually short or long durations. A statistical flag alone does not indicate a recording error, so no songs were removed on this basis.
- **Preprocessing within CV:** All preprocessing was fitted only on the training portion of each fold, then applied to its validation portion.

	count	mean	std	min	25%	50%	75%	max	missing	unique
acousticness	588.0	0.362517	0.339593	0.000001	0.03990	0.24200	0.69975	0.994	0	486
danceability	588.0	0.592745	0.173349	0.107000	0.47000	0.60450	0.71525	0.986	0	404
duration	588.0	221417.653061	69374.150420	33840.000000	185029.75000	215160.00000	246530.00000	675360.000	0	576
energy	588.0	0.594232	0.250384	0.009250	0.43325	0.62100	0.79825	0.995	0	429
instrumentalness	588.0	0.098308	0.257142	0.000000	0.00000	0.00001	0.00236	0.967	0	350
liveness	588.0	0.206297	0.178363	0.024000	0.09465	0.13100	0.27450	0.979	0	385
loudness	588.0	-8.523810	5.133955	-29.601000	-10.06750	-7.34850	-5.02650	-0.533	0	570
speechiness	588.0	0.099606	0.102676	0.023400	0.03565	0.04890	0.11650	0.570	0	403
tempo	588.0	120.167100	27.967387	59.896000	99.01750	119.96300	137.99350	204.162	0	583
valence	588.0	0.497596	0.240549	0.033200	0.29600	0.48300	0.68825	0.975	0	424
key	588.0	4.833333	3.602400	0.000000	1.00000	5.00000	8.00000	11.000	0	12
mode	588.0	0.744898	0.436290	0.000000	0.00000	1.00000	1.00000	1.000	0	2
time_signature	588.0	3.896259	0.397333	1.000000	4.00000	4.00000	4.00000	5.000	0	4

Feature Selection

Candidate Feature Sets

Numerical features: In the development data, liked songs tend to have higher acousticness and lower energy, loudness, and speechiness. danceability also shows a smaller difference. The two classes still overlap for every feature, so these patterns need to be tested together.

Categorical features: Like rates vary across key, mode, and time_signature. For example, the like rate is 78.8% for key=0 and 35.8% for key=1. Some categories contain very few songs, so these differences may be unreliable.

Feature set	Features	Question
A: Five numerical features	acousticness, danceability, energy, loudness, speechiness	Are these five features enough?
B: A + categorical features	A plus key, mode, time_signature	Do the categorical features add useful information?
C: All features	All 13 features	Do the remaining features help when combined with the others?

Candidate Hyperparameters

These are initial candidate ranges. I will select among them using five-fold cross-validation .

Model	Values to test	Reason
Logistic Regression	C: 0.01, 0.1, 1, 10	Values spaced by a factor of ten let us compare stronger and weaker regularization.
Linear SVM	C: 0.01, 0.1, 1, 10	Compare different penalties for classification errors.
Random Forest	n_estimators: 100, 300 max_features: sqrt, 0.8 min_samples_leaf: 1, 5	Test whether more trees improve stability, and how feature sampling and leaf size affect the trees.
Gradient Boosting	n_estimators: 50, 100, 200 learning_rate: 0.05, 0.1 max_depth: 1, 2	Compare shorter and longer boosting runs, smaller and larger updates, and shallow tree depths.

Five-Fold Cross-Validation

- I reserved 148 songs as a holdout set and divided the 588 development songs into five folds. Stratified splits maintained similar class proportions.
- In each round, four folds were used for training and one for validation. Every candidate combination of model, feature set, and hyperparameters used the same five-fold splits.
- Each candidate received five validation accuracy scores. Their mean was used to select the best configuration for each model and then the final model.
- Five-fold CV reduced dependence on a single validation split while keeping computation manageable.
- The final model was selected using CV only. Holdout scores were calculated afterward for comparison and were not used to change the selection.

Results

Model	Selected feature set	Selected hyperparameters	Mean CV accuracy	Holdout accuracy
Random Forest	A: five numerical features	n_estimators=300, max_features="sqrt", min_samples_leaf=5	84.01%	81.76%
Gradient Boosting	A: five numerical features	n_estimators=50, learning_rate=0.05, max_depth=2	83.84%	81.76%
Logistic Regression	C: all features	C=0.01	81.62%	81.76%
Linear SVM	C: all features	C=0.1	81.46%	76.35%

Each row shows the configuration selected for that model using the same five CV folds. Holdout scores were used for evaluation only.

Always predicting “Like” gives a holdout accuracy of 60.14% (89/148). All four models exceeded this baseline.

Comparison and Final Choice

Comparison of models. Logistic Regression and Linear SVM achieved similar CV accuracy (81.62% and 81.46%). Random Forest and Gradient Boosting scored higher (84.01% and 83.84%), suggesting that more flexible feature combinations may help on this dataset. However, Logistic Regression matched both tree ensembles on the holdout set at 81.76%, while Linear SVM achieved 76.35%. The tree ensembles therefore showed no holdout accuracy advantage over Logistic Regression.

Feature selection. Both tree ensembles performed best in CV with the five numerical features: acousticness, danceability, energy, loudness, and speechiness. Both linear models performed best with all 13 features. Adding categorical features to set A did not improve CV accuracy for any of the four models, showing that additional inputs do not necessarily improve prediction.

Final choice. I selected Random Forest because it had the highest mean CV accuracy under the chosen selection rule. It correctly classified 121 of 148 holdout songs (81.76%). Its CV lead over Gradient Boosting was only 0.17 percentage points, and three models tied on holdout accuracy. Random Forest was therefore a reasonable choice for the final predictions, although the results do not demonstrate a clear advantage over the alternatives.

Predictions for the 200 Unlabeled Songs

After selection, I refitted the Random Forest model on all 736 deduplicated labeled songs, keeping the selected features and hyperparameters fixed.

I used this model to predict the 200 songs in songs_to_classify.csv, preserving their original row order. The predictions were submitted as a 200-character string, where 0 means Dislike and 1 means Like.

The true labels are not available to me, so I cannot calculate accuracy for these songs locally. The leaderboard result was not yet available when this report was prepared. I will present and discuss it during the oral presentation.

Use of AI Tools

AI tools were used to help implement and document the code, interpret data visualizations, and explain classification methods. I reviewed the suggestions, ran the experiments, and made the final modeling decisions myself.